CSA C22.3 Overhead Line Design — Digitized Code Reference & Engineering Workflow App

Detailed Plan & Product Specification

Item	Value
Discipline	Overhead transmission & distribution (T&D) line design — Canada
Primary standards	CSA C22.3 No. 1 *Overhead Systems*; CSA C22.3 No. 60826 *Design Criteria of Overhead Transmission Lines*
Secondary standards	CSA C22.3 No. 6, No. 7, CSA O15, CSA S16/S37/A23.3/O86, CSA C411.1, CSA C83, NBCC climatic data, ASCE 10/48, IEEE 738/524/80
Audience	Line design engineers (EIT / P.Eng.), technologists, checkers, utility standards groups, software team
Status	Planning — v1.0

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>
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> ⚠️ **Warning**
>
> **Verification required.** Clause numbers, table numbers and numerical values cited in this document are *indicative* and written from general practitioner knowledge across several editions (e.g., C22.3 No. 1-15 / No. 1:20, C22.3 No. 60826-10 and later). Every value marked **[VERIFY]** — and in practice every value — must be checked by a qualified SME against the edition adopted by the governing jurisdiction **before** it is loaded into the production code library.

> ❗ **Important**
>
> **Professional responsibility.** The app is a decision-support and productivity tool. It does not replace the engineering judgment of the P.Eng. who authenticates (stamps) the design. All outputs must be traceable, reproducible and independently checkable, consistent with provincial regulator guidance (EGBC, APEGA, PEO, OIQ, etc.) on the use of engineering software and documented independent checks.

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1. Executive Summary

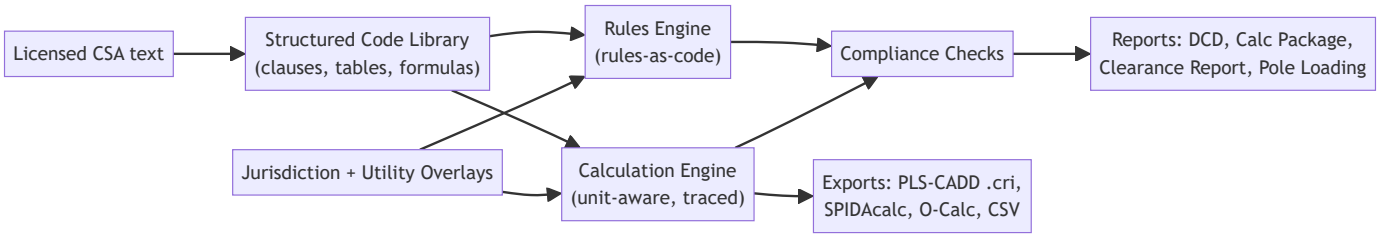
Canadian overhead line design is governed by the **CSA C22.3** series, which forms Part III of the Canadian Electrical Code and is adopted (directly or by reference, often with amendments) by provincial/territorial regulations. A line designer spends a large share of time on **non-creative, code-lookup and cross-checking tasks**:

- Identifying the loading zone, climatic parameters, return periods and construction grade.
- Translating code clauses into load cases for PLS-CADD / PLS-POLE / TOWER / SPIDAcalc / O-Calc.
- Looking up clearance tables and applying voltage adders, sag conditions and owner/jurisdiction overlays.
- Re-checking the same rules on every span, crossing and pole.
- Writing design criteria documents (DCDs) and calc packages with clause traceability.

This plan proposes two linked deliverables:

Deliverable	What it is	Primary value
A. Digitized Code Library	A structured, versioned, searchable, machine-readable representation of C22.3 No. 1 and No. 60826 (clauses, tables, formulas, figures, definitions, cross-references) plus jurisdiction & utility overlays	Instant, correct, edition-aware lookup; the single source of truth that drives calculators and rule checks
B. LineDesk CA (workflow app)	A project-based application that follows the engineer's actual workflow — from scoping to issued-for-construction — with calculators, a criteria builder, a compliance engine, integrations and report generation	40–60 % reduction in criteria/clearance/check effort (target), fewer non-conformances, consistent audit-ready documentation

The key architectural idea is **"Code-as-Data → Rules-as-Code → Calcs-with-Trace"**: every number the app produces carries a citation back to the exact edition, clause and table row it came from.



2. The CSA C22.3 Family & Regulatory Landscape

2.1 Standards in scope

Standard	Title (short)	Role in line design	Priority for digitization
CSA C22.3 No. 1	Overhead Systems	Deterministic loading, clearances, separations, strength (overload capacity factors), grounding, guys, joint use. Governs most distribution and many transmission lines.	**P1**
CSA C22.3 No. 60826	Design Criteria of Overhead Transmission Lines (Canadian adoption of IEC 60826 with Canadian deviations/annex)	Reliability-based loading (wind, ice, combined), security (failure containment) and safety (construction & maintenance) loads; strength coordination; Canadian climatic data annex.	**P1**
CSA C22.3 No. 6	Principles and practices of electrical coordination between pipelines and electric supply lines	AC interference, parallelism/crossings with pipelines	P3
CSA C22.3 No. 7	Underground Systems	Riser poles, transitions, dips	P3
CSA C22.3 No. 3 / No. 8 / No. 9	Electrical coordination / railway electrification / DER interconnection	Situational	P4

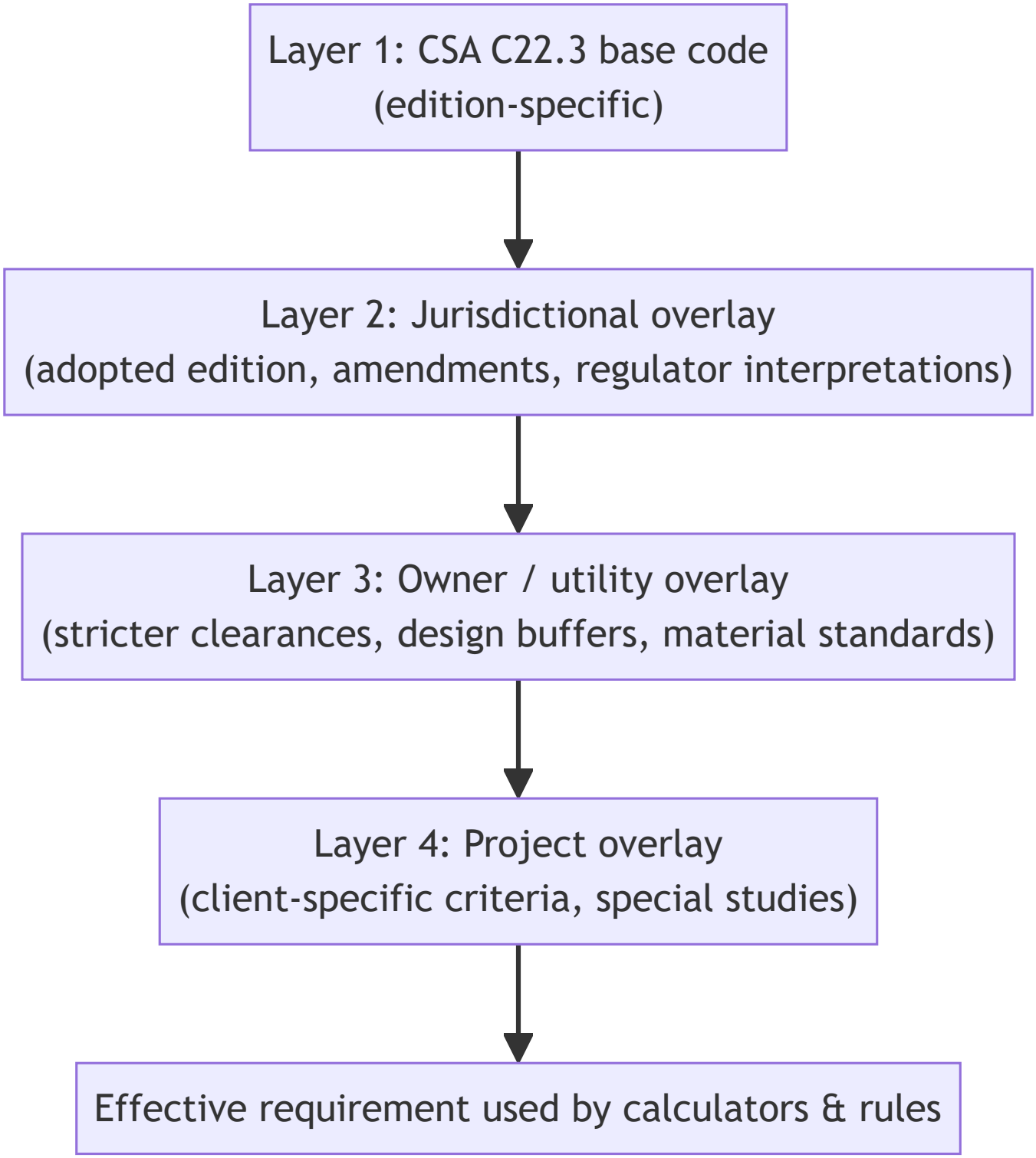
2.2 Companion standards the engineer constantly references

Topic	Standard(s)
Wood poles	CSA O15 (classes, species, dimensions, fibre stress)
Steel poles / lattice	CSA S16, CSA S37, ASCE/SEI 48 (steel poles), ASCE 10 (lattice towers)
Concrete poles / foundations	CSA A23.3, CSA A14 (concrete poles), geotechnical references (e.g., CFEM)
Wood crossarms / timber	CSA O86, utility material specs
Insulators	CSA C411.1 (and IEC 61109/60383 equivalents)
Pole line hardware	CSA C83
Climatic data	NBCC Appendix C (climatic design data), ECCC records, C22.3 No. 60826 Canadian annex
Ampacity / thermal rating	IEEE 738, CIGRE TB 601 / TB 207
Sag-tension & conductor	Aluminum Association / Southwire data, CIGRE TB 324 (sag-tension), CIGRE TB 273 (vibration / H/w)
Grounding	IEEE 80, IEEE 524 (stringing), utility standards
Railway crossings	Transport Canada *Wire Crossings and Proximities Regulations*, railway company standards

Topic	Standard(s)
Navigable waters	*Canadian Navigable Waters Act* (Transport Canada)
Aviation	Transport Canada CAR Standard 621 (obstruction marking & lighting), NAV CANADA land use submission
Pipelines	Canada Energy Regulator damage-prevention regulations, provincial pipeline acts

2.3 Regulatory adoption (overlay concept)

The code itself is not automatically law; each province/territory adopts it by regulation, sometimes with amendments, and utilities frequently adopt stricter internal standards. The app must model **three layers**:



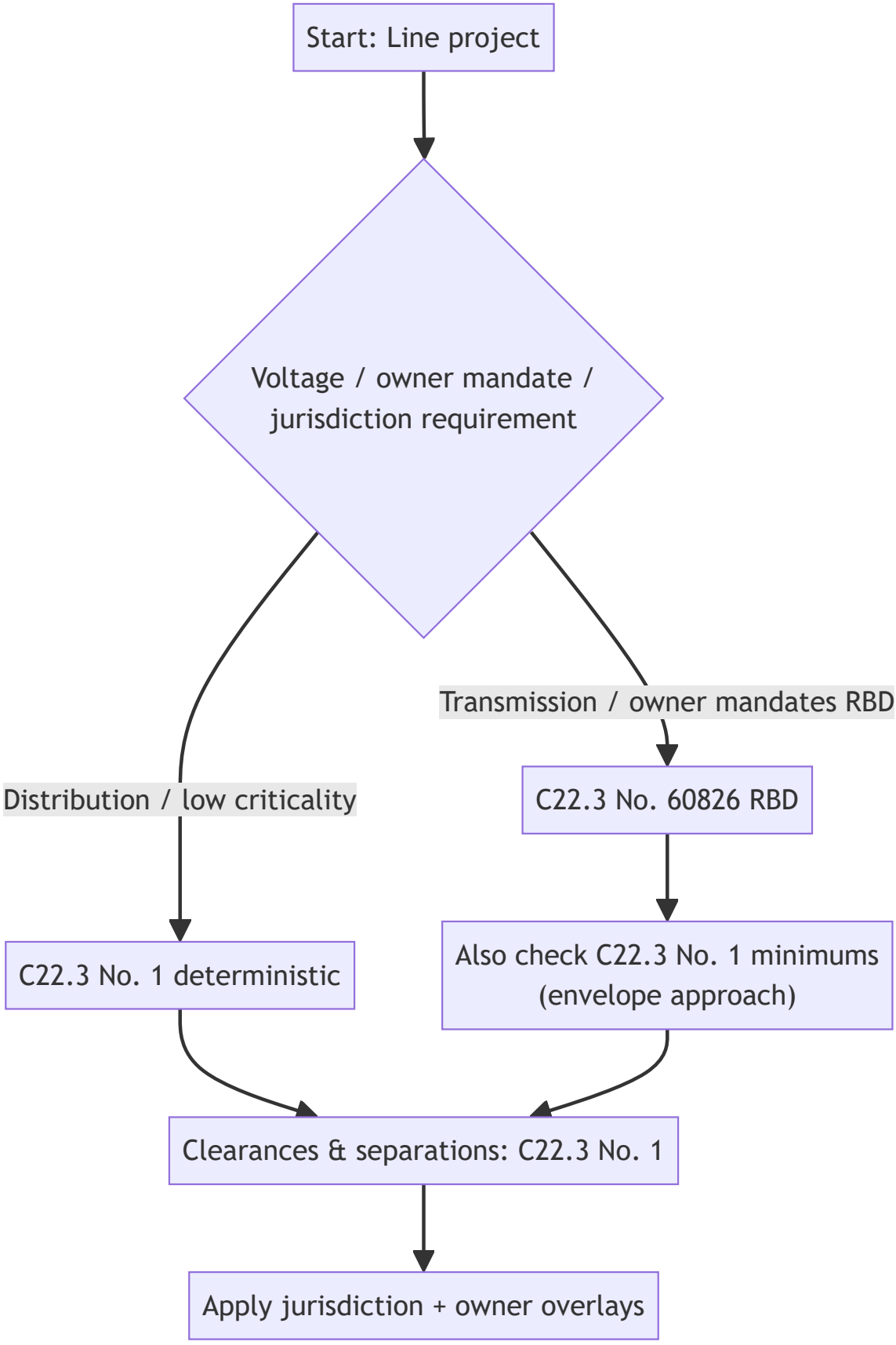
Examples to capture per jurisdiction (to be researched and verified by SMEs): adopted edition of C22.3 No. 1, whether No. 60826 is mandated for transmission above a given voltage, provincial safety-code amendments (e.g., Ontario Reg. 22/04 for distributors, Alberta Electrical Utility Code, BC Electrical Safety Regulation, Québec practice via Hydro-Québec standards), and authority-having-jurisdiction (AHJ) contacts.

> ℹ️ **Note**
>
> The **effective requirement** is always the most restrictive applicable value unless an overlay explicitly *replaces* (rather than *adds to*) the base requirement. The data model in §8 supports both `mode: restrict` and `mode: replace`, with mandatory justification text.

2.4 Deterministic vs reliability-based design path

Criterion	C22.3 No. 1 (Deterministic)	C22.3 No. 60826 (Reliability-based)
Typical use	Distribution, sub-transmission, many utility transmission lines	Transmission (typically higher voltage / critical lines), owner-mandated
Loading basis	Prescribed loading zones from map (e.g., Heavy, Medium A, Medium B, Severe… **[VERIFY]**) + temperature	Climatic return-period values (50/150/500 yr) from site data, terrain, height and span factors
Strength approach	Loads × **Overload Capacity Factors (OCFs)** ≤ rated/ultimate strength, by **construction grade**	$\gamma_U \cdot Q_T \leq \Phi_R \cdot R_c$ with strength coordination
Extra cases	Unbalanced/longitudinal, deadend, angle; special loadings	Security (broken wire, RSL), safety (construction & maintenance), failure sequence coordination
Clearances	**Always C22.3 No. 1** (clearances/separations are not replaced by 60826)	Clearances still per No. 1
Output	Load cases for structure design, sag-tension limits	Same, plus reliability documentation

Practical reality: most Canadian transmission projects check **both** and adopt the envelope (governing) case; clearances and separations always come from No. 1 (plus overlays).


Start: Line project
Voltage / owner mandate / jurisdiction requirement
Distribution / low criticality
Transmission / owner mandates RBD
C22.3 No. 1 deterministic
C22.3 No. 60826 RBD
Also check C22.3 No. 1 minimums (envelope approach)
Clearances & separations: C22.3 No. 1
Apply jurisdiction + owner overlays

3. Functional Map — CSA C22.3 No. 1 (Overhead Systems)

Rather than reproducing clause text, the digitized library maps the standard into **functional topics**, each with the *inputs the engineer must supply* and the *outputs the code provides*. Clause numbering below is indicative **[VERIFY per edition]**.

#	Topic area	Engineer's question	Code provides	Inputs needed	Output used in
3.1	Scope, definitions	"Does this apply? What does *supply conductor / communication line / structure / crossing* mean?"	Definitions, scope limits	Facility type	All
3.2	General requirements	"What are baseline obligations (accessibility, identification, materials)?"	General rules	—	DCD
3.3	**Vertical clearances above ground, rails, water**	"How high must conductors be over this road / field / rail / water?"	Clearance tables by voltage class and surface category; conditions of sag	Voltage, conductor type, crossing category, sag condition	Spotting, clearance report
3.4	**Clearances to buildings, signs, pools, bridges and other structures**	"How close can I be to this building/pool/sign, horizontally and vertically?"	Tables; wind-displaced conductor positions	Voltage, displacement (blowout)	Route & spotting
3.5	**Crossings & conflicts with other lines**	"What vertical separation is needed between my line and the line I'm crossing?"	Crossing tables by upper/lower voltage	Both voltages, temps, ice/sag	Crossing drawings
3.6	**Separations on same structure**	"Phase-to-phase, circuit-to-circuit, supply-to-communication spacing at structure and midspan?"	Tables + sag-dependent separation formula	Voltage, sag, span	Framing design
3.7	Clearance from conductors to structure / climbing space	"Do I have adequate climbing/working space and phase-to-structure clearance?"	Tables	Voltage, framing	Framing design
3.8	**Mechanical loading (loading map & conditions)**	"Which loading zone? What ice, wind, temperature?"	Loading map (figure), loading conditions table	Location	Load cases
3.9	**Construction grades & Overload Capacity Factors**	"Grade 1 or Grade 2? What OCF for vertical/transverse/longitudinal on wood/steel/concrete?"	Grade-selection criteria, OCF tables	Crossing type, voltage, material	Structure checks
3.10	**Conductor tension limits**	"Max allowable tension under loaded/unloaded conditions?"	Limits as % of rated strength	Conductor, condition	Sag-tension

#	Topic area	Engineer's question	Code provides	Inputs needed	Output used in
3.11	**Strength of structures (wood, steel, concrete, composite)**	"Is my pole/tower strong enough with OCFs and deterioration?"	Strength requirements, replacement thresholds	Structure data	Pole/tower design
3.12	**Guys and anchors**	"Guy strength, anchor capacity, guy insulators, guarding?"	Strength factors, guy insulation rules	Guy layout, voltage	Guy design
3.13	Crossarms, pins, hardware	"Hardware strength factors?"	Requirements	Hardware ratings	Framing
3.14	**Insulators**	"Mechanical strength ratio, electrical rating?"	Strength factors vs rated loads	Insulator ratings	Insulator selection
3.15	**Grounding & bonding**	"What must be grounded, how, with what resistance/conductor?"	Grounding requirements	System type (multigrounded neutral etc.)	Grounding plan
3.16	**Joint use** (supply + communications)	"Can telecom attach? At what separation? Who loads the pole?"	Separations, climbing space, attachment rules	Attachment heights	Pole loading / attachment applications
3.17	Equipment on structures	"Transformers, switches, risers: clearances & support?"	Equipment mounting rules	Equipment data	Framing
3.18	Annexes (informative)	Explanatory methods	Background	—	Reference

3.1 Example: C22.3 No. 1 loading conditions (structure of the data)

> ⚠️ **Warning**
>
> Values below are **illustrative of the data structure** and reflect commonly cited values; populate from the licensed table of the adopted edition **[VERIFY]**.

Loading condition	Radial ice (mm)	Wind pressure (Pa)	Temperature (°C)	Status
Heavy	12.5	400	−20	[VERIFY]
Medium A	6.5	400	−20	[VERIFY]
Medium B	12.5	300	−20	[VERIFY]
Severe / other zones in edition	— populate —	— populate —	— populate —	[VERIFY]

Additional C22.3 No. 1 conditions to digitize: max-temperature sag condition for clearances, ice-loaded sag condition, wind-displacement (blowout) condition for horizontal clearances, special/extreme loadings (e.g., higher-wind or heavy-snow areas, structures above a specified height), and longitudinal/unbalanced loading.

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4. Functional Map — CSA C22.3 No. 60826 (Reliability-Based Design)

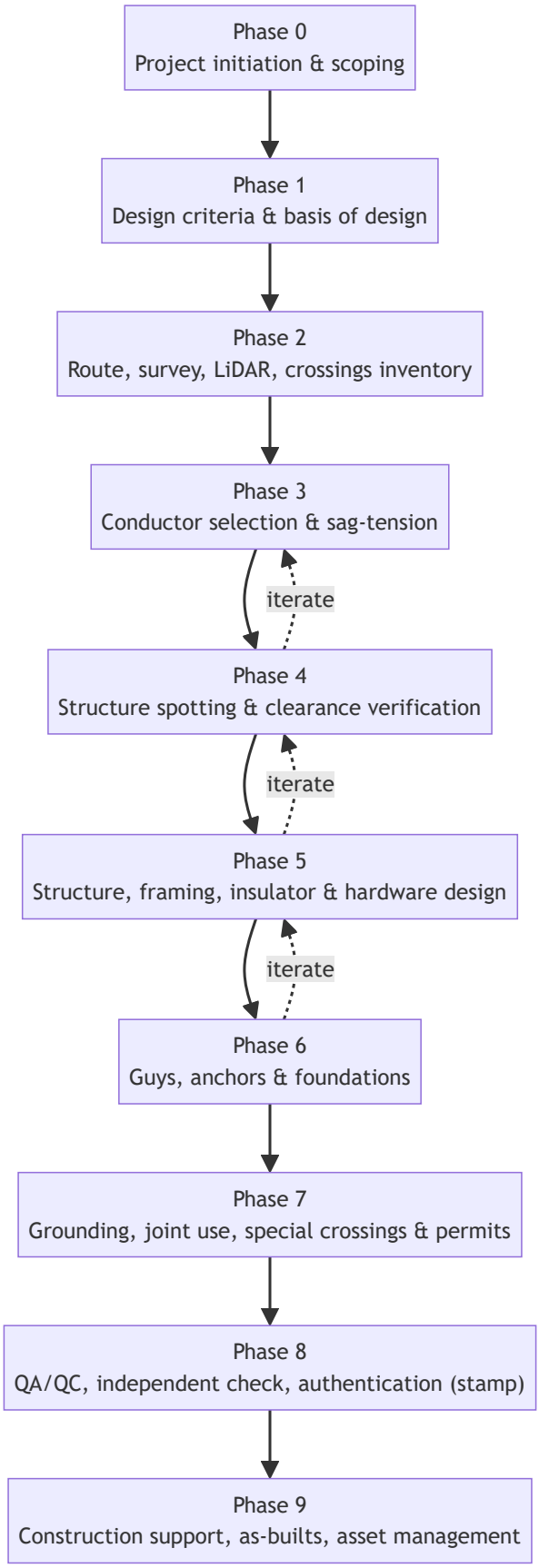
#	Topic area	Content	Engineer's decision
4.1	Design philosophy	Reliability, security, safety requirements; limit-state design equation	Choose reliability level
4.2	**Reliability levels**	Level 1/2/3 → climatic return periods **50 / 150 / 500 years**	Based on line importance, voltage, owner policy
4.3	**Climatic data**	Reference wind speed V_R (10-min mean at 10 m, terrain category B), ice thickness / mass, temperature; **Canadian annex weather data by location**	Site data source & conversion
4.4	**Wind loads**	Dynamic reference pressure, terrain roughness factor K_R, combined wind factors G_c / G_t, span factor G_L, drag coefficients C_x for conductors, insulators, supports, angle of incidence Ω	Terrain category per segment, heights
4.5	**Ice loads**	Ice types (glaze, rime, wet snow), densities, design ice mass with return period; unbalanced ice	Ice type, unbalanced scenarios
4.6	**Combined wind + ice**	High-probability ice with low-probability wind and vice versa; drag on iced conductors	Combination factors
4.7	Temperature effects	Minimum temperature, temp with wind/ice	Temps per case
4.8	**Security loads**	Torsional/longitudinal loads (broken conductor, residual static load), failure containment	Containment strategy (anti-cascade structures)
4.9	**Safety loads**	Construction & maintenance (C&M) loads, load factors for workers	C&M procedures
4.10	**Strength of components**	Characteristic strength R_c, strength factor $\Phi_R = f(\Phi_N, \Phi_Q, \Phi_S, \Phi_C)$, exclusion limits, strength coordination (failure sequence)	Which component fails first
4.11	Foundations, conductors, insulators, hardware	Component-specific strength and factors	Component selection
4.12	Annexes	Statistical methods, data conversion, Canadian data tables and deviations	Data processing

4.1 Key reference values (indicative)

Item	Value	Status
Reliability Level 1 / 2 / 3 return period	50 / 150 / 500 yr	Widely cited (IEC basis) — [VERIFY Canadian deviation]
Terrain roughness K_R: A / B / C / D	1.08 / 1.00 / 0.85 / 0.67	IEC basis — [VERIFY]
Air density μ	1.225 kg/m³ (15 °C, sea level)	Standard
Glaze ice density	900 kg/m³	Standard
Drag coefficient, conductors C_{xc}	1.00 (typical)	[VERIFY]
Drag coefficient, insulators C_{xi}	1.20 (typical)	[VERIFY]

5. The Engineer's End-to-End Workflow

5.1 Overview


Phase 0
Project initiation & scoping
Phase 1
Design criteria & basis of design
Phase 2
Route, survey, LiDAR, crossings inventory
Phase 3
Conductor selection & sag-tension
iterate
Phase 4
Structure spotting & clearance verification
iterate
Phase 5
Structure, framing, insulator & hardware design
iterate
Phase 6
Guys, anchors & foundations
Phase 7
Grounding, joint use, special crossings & permits
Phase 8
QA/QC, independent check, authentication (stamp)
Phase 9
Construction support, as-builts, asset management

Each phase below lists: **Objective** → **Code touchpoints** → **Inputs** → **Outputs** → **Pain points** → **Automation opportunities**.

Phase 0 — Project initiation & scoping

Aspect	Detail
Objective	Define facility type, voltage, circuits, jurisdiction, owner, schedule, code path
Code touchpoints	C22.3 No. 1 scope/definitions; jurisdictional adoption; decision on No. 60826
Inputs	Client scope, system studies (load flow, fault levels), route corridor, owner standards
Outputs	Project charter, applicable-codes register, design-path decision
Pain points	Wrong edition used; missed provincial amendments; owner standard conflicts discovered late
Automation	**Project setup wizard**: location + voltage + owner → auto-generated *Applicable Codes & Standards Register* with adopted editions and overlays; flags conflicts

Phase 1 — Design criteria & basis of design (the most code-intensive phase)

Aspect	Detail
Objective	Produce the **Design Criteria Document (DCD)** that governs every downstream calc
Code touchpoints	No. 1 loading map & conditions, construction grade, OCFs, tension limits, clearance conditions; No. 60826 reliability level, climatic data, wind/ice/combined loads, security & safety loads, strength factors
Inputs	Coordinates/route, elevation, terrain, historical climate (Canadian annex, NBCC, ECCC), owner experience (ice storms, galloping), conductor candidates
Outputs	DCD; load case matrix; temperature/tension limits; clearance criteria table with buffers; PLS-CADD criteria file
Pain points	Manual map reading; inconsistent load case naming; transcription errors into PLS-CADD; poor traceability
Automation	**Criteria Builder**: map click / KMZ import → loading zone(s), climatic values interpolated, return-period scaling, terrain-based factors → **auto load case matrix** (No. 1 + 60826 envelope) → export to `.cri`, SPIDAcalc, O-Calc; DCD auto-drafted with clause citations

Typical DCD content checklist (see Appendix B for template): design code path & editions; reliability level; loading zone(s); weather cases (ice, wind, temp); construction grade by location; OCFs / strength factors; conductor tension limits (initial/final, every-day/H-w); clearance conditions and buffers; insulator swing cases; galloping criteria; security/containment; C&M loads; structure families; material standards.

Phase 2 — Route, survey, LiDAR & crossings inventory

Aspect	Detail
Objective	Build terrain model and identify every feature that drives clearances
Code touchpoints	No. 1 surface categories (roads, driveways, fields, rail, water, pedestrians-only areas), buildings/pools/signs, crossings with other utilities

Aspect	Detail
Inputs	LiDAR point cloud, orthophoto, GIS (roads, rail, parcels, pipelines, water bodies), field reconnaissance
Outputs	Feature-coded terrain (PLS-CADD feature codes), crossings register, permits register
Pain points	Classifying thousands of features by code category; missing crossings; inconsistent feature codes
Automation	**Feature-to-code-category mapper**: GIS/LiDAR classification → C22.3 surface category with required clearance auto-assigned; crossings register auto-generated with AHJ/permit requirements (rail, navigable water, highway, pipeline, aviation)

Phase 3 — Conductor selection & sag-tension

Aspect	Detail
Objective	Select conductor/OHGW/OPGW and establish stringing tensions
Code touchpoints	No. 1 tension limits; No. 60826 conductor strength; ice/wind cases from DCD
Inputs	Ampacity requirement (IEEE 738), conductor library, ruling spans, creep, vibration limits (H/w)
Outputs	Sag-tension tables (initial/final, creep/load), stringing charts
Pain points	Re-running cases when criteria change; checking every limit manually
Automation	**Sag-tension engine** with automatic limit checks against DCD; "governing case" identification; stringing chart generator

Phase 4 — Structure spotting & clearance verification

Aspect	Detail
Objective	Locate structures, verify every clearance and separation at every critical condition
Code touchpoints	No. 1 vertical clearances, horizontal clearances (blowout), crossings, midspan separations, clearances to structures; overlays/buffers
Inputs	Terrain model, structure families, sag-tension, DCD clearance table
Outputs	Plan & profile drawings, clearance report, crossing drawings
Pain points	Long clearance reports reviewed manually; voltage adders/buffers applied inconsistently
Automation	**Clearance compliance engine**: ingest PLS-CADD clearance/violation reports or computed catenaries → compare to effective requirement per feature → violations list with clause citation, margin and governing condition

Phase 5 — Structure, framing, insulator & hardware design

Aspect	Detail
Objective	Confirm structural adequacy of poles/towers, crossarms, insulators and hardware
Code touchpoints	No. 1 grades & OCFs, strength of wood/steel/concrete, insulator & hardware strength factors, separations at structure; No. 60826 strength factors & coordination

Aspect	Detail
Inputs	Load cases, structure geometry, material properties (CSA O15 etc.)
Outputs	Structure usage charts (allowable span/angle), pole class selection, framing drawings, insulator swing charts
Pain points	Iterating pole classes; usage chart regeneration; swing checks
Automation	**Pole/structure check calculators**, **usage-chart generator** (allowable wind/weight span vs line angle per structure family), **insulator swing calculator** with clearance envelopes

Phase 6 — Guys, anchors & foundations

Aspect	Detail
Objective	Size guys, anchors and foundations
Code touchpoints	No. 1 guy/anchor strength factors, guy insulators & guards; No. 60826 foundation strength factors
Inputs	Structure reactions, geotechnical data, anchor products
Outputs	Guy/anchor schedule, embedment depths, foundation designs
Automation	**Guy & anchor calculator**, direct-embed calculator (with geotechnical inputs), auto-generated staking sheet entries

Phase 7 — Grounding, joint use, special crossings & permits

Aspect	Detail
Objective	Complete grounding plan, joint-use attachment design, crossing approvals
Code touchpoints	No. 1 grounding & bonding, joint-use separations & climbing space; No. 6 (pipelines); external regs (rail, water, aviation)
Outputs	Grounding plan, attachment approvals, permit packages
Automation	**Joint-use attachment checker** (separation + pole loading in one pass), **permit package generator** with required drawings/clearance tables

Phase 8 — QA/QC, independent check, authentication

Aspect	Detail
Objective	Independent review, resolution, stamping
Outputs	Checked calc package, comment log, stamped drawings/DCD
Pain points	Checker must re-derive inputs; no single trace from number → clause
Automation	**Calc trace & checker mode**: every output shows inputs, formula, clause, edition, overlay; checker signs off per item; immutable revision history

Phase 9 — Construction support, as-builts & asset management

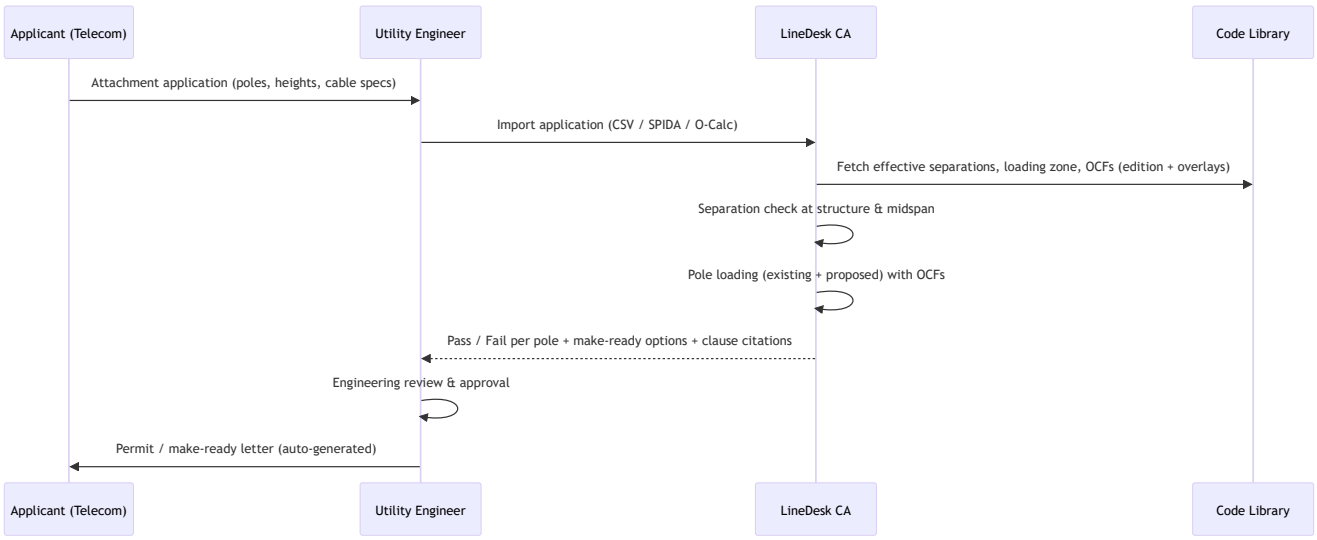
Aspect	Detail
Objective	RFIs, field changes, as-builts, life-cycle assessments (pole deterioration, uprating)
Code touchpoints	No. 1 strength of deteriorated structures/replacement criteria; clearance re-verification after changes
Automation	**Field mode** (offline clearance & loading lookup, measurement entry), **re-rating / uprating workflow** (new temp → re-check clearances), **residual-strength pole assessment** using inspection data

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5.2 Workflow variants (common engagement types)

Variant	Typical frequency	Key code-driven steps	Highest-value automation
D1 — Distribution pole-attachment application (telecom/joint use)	Very high (thousands/yr)	Separation at structure & midspan, pole loading with new attachment, guying	Batch pole loading + separation check; auto approve/deny with reasons
D2 — New distribution feeder / subdivision	High	Loading zone, grade, clearances to ground/buildings, standard framing	Standard-unit design + auto clearance checks
D3 — Pole replacement / storm hardening	High	Strength, OCFs, deterioration	Rapid class selection, storm-hardening scenarios
T1 — New transmission line	Medium	Full Phase 0–9, No. 60826 + No. 1 envelope	Criteria builder, clearance engine, usage charts
T2 — Line uprating (thermal)	Medium	Higher max temp → clearances, sag	Batch re-check, mitigation suggestions (raise, re-sag, reconductor)
T3 — Rebuild / reconductoring	Medium	Loads change, structure re-check	Delta analysis vs existing
X — Third-party proximity requests (buildings, cranes, pipelines)	High	Clearances to buildings/structures, No. 6	Quick proximity calculator + letter generator

5.3 Example: Distribution pole-attachment flow (D1)



6. Core Engineering Calculation Library

All calculators share: **unit-aware inputs (SI primary, imperial display optional), validation ranges, explicit code-path tag (No. 1 / No. 60826 / both), and a calc-trace record**. Formulas below are standard engineering relations; code-specific coefficients come from the digitized library at runtime.

6.1 Weather & loading

Dynamic wind pressure (general):

$$q = \tfrac{1}{2}\,\rho\,V^2 \qquad (\rho = 1.225\ \mathrm{kg/m^3} \Rightarrow q \approx 0.613\,V^2\ [\mathrm{Pa}],\ V \text{ in m/s})$$

No. 60826 dynamic reference pressure:

$$q_0 = \tfrac{1}{2}\,\tau\,\mu\,(K_R\,V_R)^2$$

where τ = air density correction (temperature/altitude), μ = 1.225 kg/m³, K_R = terrain roughness factor, V_R = reference wind speed for the selected return period.

Wind load on conductors (No. 60826 form):

$$A_c = q_0 \cdot C_{xc} \cdot G_c \cdot G_L \cdot d \cdot L \cdot \sin^2\Omega$$

Radial glaze ice weight per unit length (ρ_ice = 900 kg/m³, t and d in mm):

$$w_{ice} = \rho_{ice}\,g\,\pi\,t\,(t+d) \times 10^{-6} \approx 0.02774\,t\,(t+d)\ \ [\mathrm{N/m}]$$

Iced diameter: $D = d + 2t$

Resultant unit load (deterministic, ice + wind):

$$w_r = \sqrt{(w_c + w_{ice})^2 + (q\,D)^2}$$

(Any code-specified additive constant or temperature association must be applied per the adopted edition **[VERIFY]**.)

Return-period scaling: values from the Canadian annex for the base return period scaled to the selected reliability level using the code's conversion factors (table-driven, interpolated) **[VERIFY method per edition]**.

6.2 Sag-tension

Relation	Formula
Ruling (equivalent) span	$L_R = \sqrt{\dfrac{\sum L_i^3}{\sum L_i}}$
Parabolic sag	$S = \dfrac{wL^2}{8H}$

Relation	Formula
Catenary sag	$S = \frac{H}{w}\left[\cosh\left(\frac{wL}{2H}\right) - 1\right]$
Conductor length (catenary)	$\ell = \frac{2H}{w}\sinh\left(\frac{wL}{2H}\right)$
Change of state	Strain summation: elastic + thermal + creep (initial/final, load-history), solved iteratively
Tension limit check	$T_{max,case} \le k_{case} \cdot RTS$ with k from code/DCD
Vibration (every-day)	H/w (catenary parameter) vs CIGRE TB 273 limits

6.3 Clearances & separations

Calculation	Approach
Vertical clearance	Required = table value(voltage class, surface category) + voltage adder (if applicable) + owner buffer; Actual = conductor elevation at governing condition (max temp / ice) − ground/feature elevation
Horizontal clearance (blowout)	Conductor swing under code wind condition: $\phi = \arctan\left(\frac{q\,D}{w_c + w_{ice}}\right)$; displaced position vs building/feature
Insulator swing	$\theta = \arctan\left(\frac{q\,D\,L_{wind} + P_{ins}/2}{w\,L_{weight} + W_{ins}/2}\right)$ vs required air gap per swing case
Midspan separation	Code sag-dependent formula (voltage + sag terms) **[VERIFY]**, plus galloping ellipse checks where required by owner
Crossing separation	Upper line at max sag condition vs lower line at min sag condition; table requirement by both voltages

6.4 Structures

Calculation	Formula / method
Groundline moment (pole)	$M_{GL} = \sum F_{c,i}\,h_i + F_{pole}\,\bar{h} + M_{P\Delta}$
Wind on pole	$F_{pole} = q\,C_d\,A_{proj}$ (tapered pole area)
Wood pole section modulus (from circumference C)	$S = \frac{C^3}{32\pi^2}$
Wood pole stress check	$\mathrm{OCF} \times M \le f_b \cdot S$ (deterministic) or $\gamma_U M \le \Phi_R R_c$ (RBD)
Guy tension	$T = \frac{H}{\sin\beta}$ (β = angle between guy and pole) × OCF ≤ rated strength × strength factor
Guyed pole buckling	Euler/Gere–Carter style tapered-column buckling check
Line-angle loads	$F_{angle} = 2T\sin(\alpha/2)$
Deadend / unbalanced	Longitudinal = full or residual tension per code case
Lattice / steel pole	Member/section checks per ASCE 10 / ASCE 48 / CSA S16 (export loads to PLS-TOWER/PLS-POLE)

6.5 Foundations & anchors

Direct-embed depth (rule of thumb + geotechnical method, e.g., Brinch Hansen / Broms for lateral), anchor capacity from manufacturer data × strength factor, rock/soil class selection — always with geotechnical sign-off inputs.

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7. Automation Strategy

7.1 Automation maturity levels

Level	Name	Description	Example
L0	Digital reading	Searchable licensed PDF	Today's baseline
L1	Structured lookup	Clauses/tables as data, filters, cross-refs	"Show vertical clearance for 25 kV over a road"
L2	Parametric calculators	Table lookups + formulas with trace	Wind/ice load, sag, swing, pole stress
L3	Criteria generation	Project-level, multi-case generation & export	DCD + PLS-CADD criteria auto-generated
L4	Compliance automation	Batch checking of models/reports against rules	Whole-line clearance & pole loading pass/fail
L5	Assisted design	Optimization & recommendations with human approval	Suggest pole class/height, structure spotting optimization, mitigation options for violations

7.2 Automation opportunities — value vs effort

#	Automation	Value	Effort	Priority
A1	Structured code library with search, cross-refs and edition switching	High	Medium	**MVP**
A2	Clearance lookup with voltage adders & overlays	Very high	Low	**MVP**
A3	Loading zone / climatic data lookup from map location	Very high	Medium	**MVP**
A4	Load case matrix generator (No. 1 + No. 60826 envelope)	Very high	Medium	**MVP**
A5	PLS-CADD criteria (.cri) / SPIDAcalc / O-Calc load case export	Very high	Medium–High	Phase 2
A6	Wood pole loading calculator with OCFs & CSA O15 data	High	Medium	**MVP**
A7	Sag-tension engine with limit checks	High	High	Phase 2
A8	Clearance report ingestion & compliance engine	Very high	Medium	Phase 2
A9	Insulator swing & blowout calculators	Medium	Low	Phase 2
A10	Joint-use batch attachment checker	Very high (distribution)	Medium	Phase 2
A11	DCD and calc-package report generator with citations	High	Medium	Phase 2
A12	Usage chart generator per structure family	High	Medium	Phase 3

#	Automation	Value	Effort	Priority
A13	Crossing register & permit package generator	Medium	Medium	Phase 3
A14	Edition diff & impact analysis ("what changes if we move to the new edition?")	High (standards groups)	Medium	Phase 3
A15	AI assistant: natural-language Q&A over licensed code with mandatory clause citations	High	Medium	Phase 3
A16	Mitigation recommender (raise structure, re-sag, reframe)	Medium	High	Phase 4
A17	Spotting optimization	Medium	Very high	Phase 4 (or rely on PLS-CADD Optimum Spotting)

7.3 Automation principles

1. **Never hide the code.** Every automated value links to clause/table/edition and overlay.
2. **Human-in-the-loop for judgment.** Automation proposes; the engineer accepts, overrides (with mandatory reason), and signs.
3. **Deterministic & reproducible.** Same inputs + same library version ⇒ identical outputs (library snapshots pinned per project).
4. **Conservative defaults.** When data is ambiguous (e.g., point on loading-zone boundary), default to the more severe value and flag.
5. **Integrate, don't replace.** PLS-CADD, PLS-POLE/TOWER, SPIDAcalc, O-Calc remain the analysis engines; LineDesk generates their inputs and checks their outputs.
6. **AI is retrieval-grounded only.** Any LLM feature answers *only* from the licensed library, cites clauses, and refuses when not grounded.

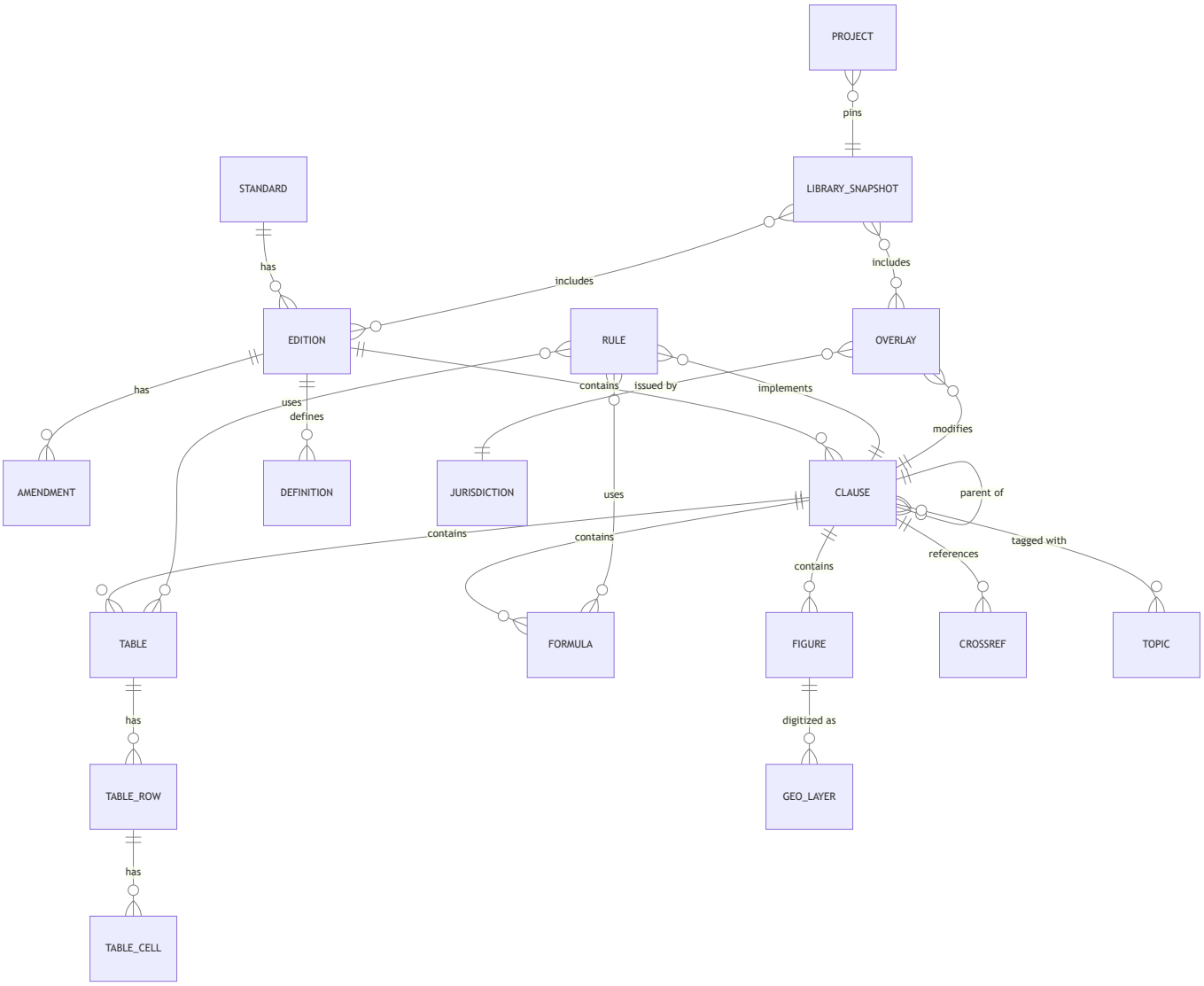
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8. The Digitized Code ("Code-as-Data")

8.1 Goals

- Faithful, versioned, machine-readable representation of each adopted edition.
- Tables are **queryable data**, not images.
- Formulas are **executable expressions** with units.
- Cross-references are **links** (graph).
- Overlays (jurisdiction, owner, project) are **separate layers**, never edits to the base.
- Every record carries **provenance** (edition, amendment, page, entered by, verified by, date).

8.2 Data model



8.3 Record schemas (illustrative)

Clause

```
{
  "id": "C22.3-1:2020/5.x.x",
  "standard": "CSA C22.3 No. 1",
  "edition": "2020",
  "amendment": null,
  "number": "5.x.x",
  "title": "Vertical clearances of conductors above ground or rails",
  "text_licensed": "<licensed clause text — only if license permits>",
  "intent_summary": "Sets minimum vertical clearance of conductors above ground/rails by voltage and surface category under specified sag conditions.",
  "normative": true,
  "topics": ["clearance.vertical", "crossing.road", "crossing.rail"],
  "applies_to": { "facility": ["supply", "communication"], "voltage_kV": [0, 1000] },
  "tables": ["C22.3-1:2020/T-clearance-vertical"],
  "formulas": [],
  "cross_refs": ["C22.3-1:2020/5.y (sag conditions)", "C22.3-1:2020/3 (definitions: 'crossing')"],
  "provenance": { "page": 0, "entered_by": "jdoe", "verified_by": "asmith_peng", "verified_on": "2026-xx-xx" },
  "status": "VERIFIED"
}
```

Table (queryable)

```
id: C22.3-1:2020/T-clearance-vertical
title: Minimum vertical clearance above ground or rails
dimensions:
  - key: surface_category       # e.g., highway, road, driveway, field, rail, pedestrian-only, water
    type: enum
  - key: voltage_class          # e.g., comm, 0-0.75kV, 0.75-22kV, 22-50kV, ...
    type: range_kV
value:
  key: clearance
  unit: m
interpolation: none             # none | linear | step
adders:
  - rule_id: R-VADDER-01        # voltage adder above a threshold [VERIFY]
rows:
  - { surface_category: road, voltage_class: "0.75-22", clearance: "<VALUE> [VERIFY]" }
notes:
  - "Applies at the governing sag condition defined in clause 5.y [VERIFY]"
status: DRAFT
```

Formula (executable)

```
id: C22.3-60826/F-q0
expression: "0.5 * tau * mu * (K_R * V_R)^2"
variables:
  tau: { unit: "1", source: "table:air_density_correction" }
  mu:  { unit: "kg/m^3", default: 1.225 }
  K_R: { unit: "1", source: "table:terrain_roughness" }
  V_R: { unit: "m/s", source: "input|climatic_data" }
result: { symbol: q0, unit: "Pa" }
tests:
  - inputs: { tau: 1.0, mu: 1.225, K_R: 1.0, V_R: 30 }
    expected: 551.25
    tolerance: 0.01
```

Rule (rules-as-code)

```
id: R-CLR-VERT-001
title: Vertical clearance to ground/feature
implements: C22.3-1:2020/5.x.x
inputs: [voltage_kV, surface_category, conductor_elevation_m, feature_elevation_m, condition]
logic: |
  required = lookup(T-clearance-vertical, surface_category, voltage_class(voltage_kV))
           + adder(R-VADDER-01, voltage_kV)
           + overlay_buffer(owner, "clearance.vertical")
  actual   = conductor_elevation_m - feature_elevation_m
  pass     = actual >= required
outputs: [required, actual, margin = actual - required, pass]
severity_on_fail: VIOLATION
citation_format: "{standard} {edition} cl. {clause}, {table} row {row}"
```

Overlay

```
id: OVL-UTILITYX-CLR-BUF
issuer: "Utility X Distribution Standards"
type: owner
mode: restrict              # restrict = add/take max ; replace = supersede (needs justification)
target: "topic:clearance.vertical"
change: { add_m: 0.30 }
justification: "Survey/model accuracy & future re-sag buffer"
effective: 2026-01-01
```

8.4 Topic taxonomy (tagging)

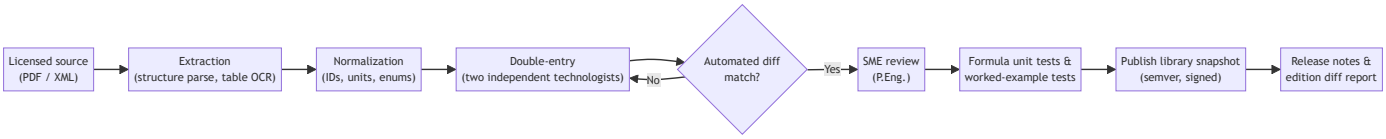
```
loading.zone | loading.ice | loading.wind | loading.temperature | loading.combined
loading.security | loading.safety | loading.longitudinal
strength.ocf | strength.grade | strength.wood | strength.steel | strength.concrete
strength.guy | strength.anchor | strength.insulator | strength.hardware | strength.foundation
conductor.tension_limit | conductor.sag_condition | conductor.vibration
clearance.vertical | clearance.horizontal | clearance.building | clearance.pool
clearance.sign | clearance.bridge | clearance.structure | clearance.water | clearance.rail
separation.same_structure | separation.midspan | separation.crossing | separation.comm_supply
grounding | bonding | jointuse | equipment | definitions
```

8.5 Figures & maps

- **Loading map** digitized into a **GIS polygon layer** (GeoJSON / vector tiles) with zone attribute and edition. Boundary uncertainty buffer (e.g., ±X km) triggers a "near boundary — conservative zone applied" flag.

- **Climatic data tables** (Canadian annex, NBCC) stored as point datasets with spatial interpolation rules exactly as the code/user-approved method prescribes (nearest station vs. interpolation must be an explicit, recorded choice).
- Figures that are diagrams (e.g., clearance envelopes) are stored as SVG with parametric overlays.

8.6 Digitization pipeline



Quality gates: 100 % double-entry match for tables; every formula has ≥ 2 test vectors; every rule has pass/fail test cases; SME sign-off recorded per clause; snapshot is immutable and hash-signed.

8.7 Edition & amendment management

- Each edition is a separate namespace (`C22.3-1:2015` , `C22.3-1:2020` , …).
- Clause **lineage links** (`supersedes` , `renumbered_from`) enable automatic **edition diff** reports: changed values, new clauses, deleted clauses, and **projects impacted**.
- Projects pin a **library snapshot**; upgrading a project is an explicit action producing an impact report.

8.8 Reference UI for the digitized code

Feature	Description
Clause browser	Tree navigation, breadcrumbs, normative/informative badges
Smart search	Full-text + topic facets + voltage/surface filters + synonyms ("ground clearance" → clearance.vertical)
Interactive tables	Pick dimensions → highlighted cell → "Use in calculator"
Live formulas	Rendered math + inline calculator + unit switching
Cross-ref graph	Hover previews, "referenced by" panel
Definitions	Inline tooltips on defined terms
Overlay view	Toggle base ↔ effective (with jurisdiction/owner deltas highlighted)
Edition compare	Side-by-side with changes highlighted
Annotations	Personal and team notes, interpretations, RFI/AHJ responses linked to clauses
Bookmarks & "cheat sheets"	Custom quick-reference cards (e.g., "25 kV distribution clearances, Utility X") exportable to PDF for field use

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9. The App — "LineDesk CA"

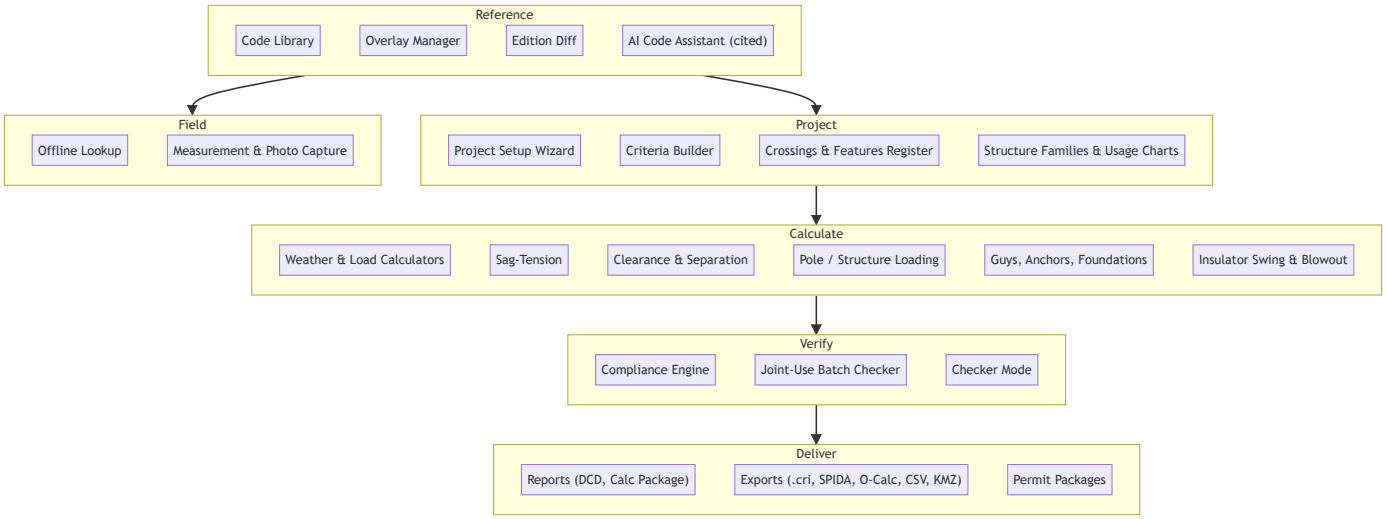
9.1 Personas

Persona	Goals	Key modules
Transmission design engineer (P.Eng.)	Fast, defensible criteria; envelope load cases; clean PLS-CADD inputs	Criteria Builder, Code Library, Compliance, Reports
Distribution design technologist	Rapid standard designs, pole loading, clearances	Calculators, Pole Loading, Joint Use
Joint-use / attachments engineer	Process high volume of attachment applications	Batch Joint-Use Checker
Checker / reviewer	Verify inputs and citations quickly	Checker Mode, Calc Trace
Standards engineer (utility)	Maintain overlays, evaluate new editions	Overlay Manager, Edition Diff
Field engineer / inspector	Offline lookups, measurements, RFIs	Field Mode (mobile)

9.2 Product principles

1. Workflow-first navigation (mirrors Phases 0–9).
2. Every number is clickable → shows its trace.
3. Offline-capable reference and calculators.
4. Import/export with industry tools is first-class.
5. Audit trail by default.

9.3 Module map



9.4 Module specifications

M1 — Project Setup Wizard

Step	Inputs	System actions	Output
1. Basics	Name, client/owner, facility type (T/D/joint), voltage(s), circuits	Validate	Project record
2. Location	Map draw / KMZ / shapefile of route	Determine province(s), loading zone(s), climatic stations, terrain hints	Location profile
3. Code path	Suggested (No. 1 / No. 60826 / envelope), editable	Pull adopted editions & overlays	Applicable Codes Register
4. Owner standards	Select owner overlay	Merge overlay	Effective rule set
5. Snapshot	Confirm	Pin library snapshot	Project ready

M2 — Criteria Builder

- **Weather panel:** loading zone(s) per segment; No. 60826 reliability level; V_R, ice, temperatures (source shown: annex station, NBCC, ECCC, owner study); return-period conversion; terrain category per segment; altitude correction.
- **Load case matrix:** auto-generated rows (No. 1 cases + No. 60826 climatic, security, safety cases), columns: ice, wind, temp, OCFs / load factors, strength factors, applicable components. Engineer can add/disable with reason.
- **Tension limits & sag conditions:** per conductor type and condition.
- **Clearance criteria table:** per feature category and voltage — base + adder + owner buffer = design value.
- **Outputs:** DCD (DOCX/PDF), `.cri` / criteria export, JSON for API.

Example generated load case matrix (illustrative):

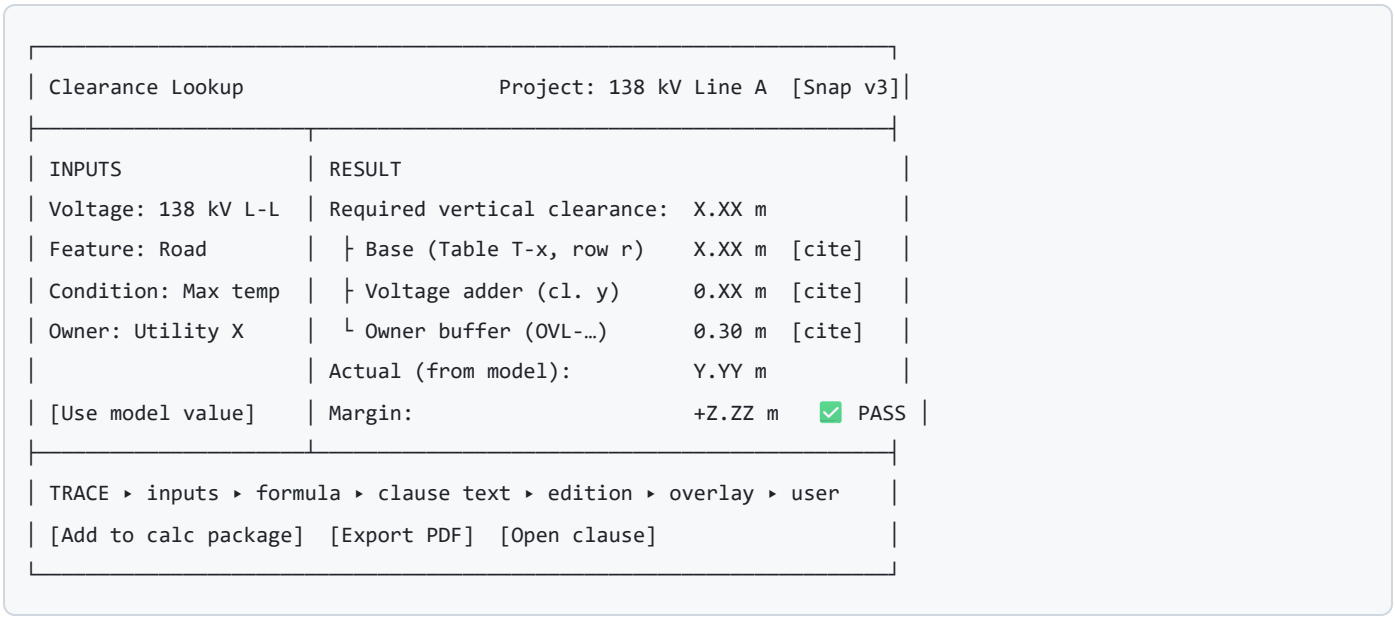
ID	Source	Description	Ice (mm)	Wind	Temp (°C)	Factors	Use
N1-HVY	No. 1	Zone loading (e.g., Heavy)	12.5 [VERIFY]	400 Pa [VERIFY]	−20	OCF by grade/material	Strength
N1-MAXT	No. 1	Max conductor temperature	0	0	T_max (rating)	—	Vertical clearance
N1-ICE	No. 1	Ice-loaded sag	per zone	0	per edition	—	Vertical clearance
N1-BLOW	No. 1	Wind displacement	0	per edition	per edition	—	Horizontal clearance
R-W	60826	Reliability wind (T yr)	0	q0(V_R,T)	coincident	γ_U, Φ_R	Strength
R-I	60826	Ice without wind (T yr)	t_T	0	−5 typ. [VERIFY]	γ_U, Φ_R	Strength

ID	Source	Description	Ice (mm)	Wind	Temp (°C)	Factors	Use
R-IW1	60826	Low-prob ice + high-prob wind	t_T	reduced V	coincident	γ_U, Φ_R	Strength
R-IW2	60826	High-prob ice + low-prob wind	reduced t	V_iH	coincident	γ_U, Φ_R	Strength
R-UI	60826	Unbalanced ice	per code	0	—	—	Longitudinal
R-SEC	60826	Security: broken conductor / RSL	—	—	—	—	Containment
R-CM	60826	Construction & maintenance	—	—	—	safety factors	Safety

M3 — Calculators (stand-alone or project-linked)

Calculator	Inputs	Outputs	Code links
Weather load	Location, return period, terrain, height, conductor d	q0, ice t, g_R, iced D, resultant loads per case	No. 60826 wind/ice; No. 1 loading table
Clearance lookup	Voltage, surface/feature type, condition, owner	Required clearance with breakdown	No. 1 clearance tables + overlays
Separation	Voltages, sag, span, configuration	Required separation at structure & midspan	No. 1 separation tables/formula
Sag-tension	Conductor, spans, criteria, creep	Sag/tension tables, governing limit	No. 1 tension limits; DCD
Pole loading	Pole species/class/height, embedment, attachments, guys, angle, zone	Groundline moment, % utilization, required class	No. 1 OCFs/grade; CSA O15
Guy & anchor	Pole reactions, geometry, soil	Guy size/count, anchor type, utilization	No. 1 guy/anchor rules
Insulator swing	Wind/weight spans, insulator data, cases	Swing angles, clearance envelope plot	No. 1 / owner swing cases
Blowout	Conductor, span, wind case	Horizontal displacement, clearance to building	No. 1 horizontal clearance
Crossing	Upper/lower line data	Separation required vs actual	No. 1 crossing tables
Galloping (owner criteria)	Conductor, span, sag	Ellipse dimensions & phase interference	Owner / CIGRE method

Every calculator screen layout:



M4 — Compliance Engine

- **Inputs:** PLS-CADD clearance/violation reports, structure usage reports, PLS-POLE/TOWER results, SPIDAcalc/O-Calc results, CSV/LAS-derived clearance tables.
- **Processing:** map each record to feature category & condition → resolve effective rule → evaluate → classify (PASS / MARGINAL (< buffer) / VIOLATION / NEEDS-DATA).
- **Outputs:** interactive table + map/profile view; filter by severity, rule, structure; export clearance report with citations; open-issues list synced to checker comments.

M5 — Joint-Use Batch Checker (distribution)

- Bulk import of applications (CSV/SPIDA/O-Calc JSON).
- For each pole: separations (supply ↔ comm, comm ↔ comm, climbing space), midspan sag separations, pole loading with new attachment, guying adequacy.
- Auto-generated make-ready scope (raise, lower, transfer, replace pole with class/height).
- Approval letter templates with clause citations.

M6 — Reports & Deliverables

Report	Contents
Design Criteria Document	Auto-drafted from Criteria Builder, clause citations, overlay table, revision block
Calculation package	Selected calculations with full trace, cover sheet, checker sign-off
Clearance report	Compliance engine results by span/feature
Pole loading report	Per pole utilization, governing case
Crossing/permit package	Crossing drawings data, clearance tables, AHJ-specific forms
Codes register	Applicable standards, editions, amendments, overlays

M7 — Checker Mode

- Side-by-side "designer value vs checker independent value" for spot-checks.
- Item-level status: Not checked / Agreed / Comment / Resolved.
- Locks the revision once approved; stamps metadata (who/when/snapshot hash).

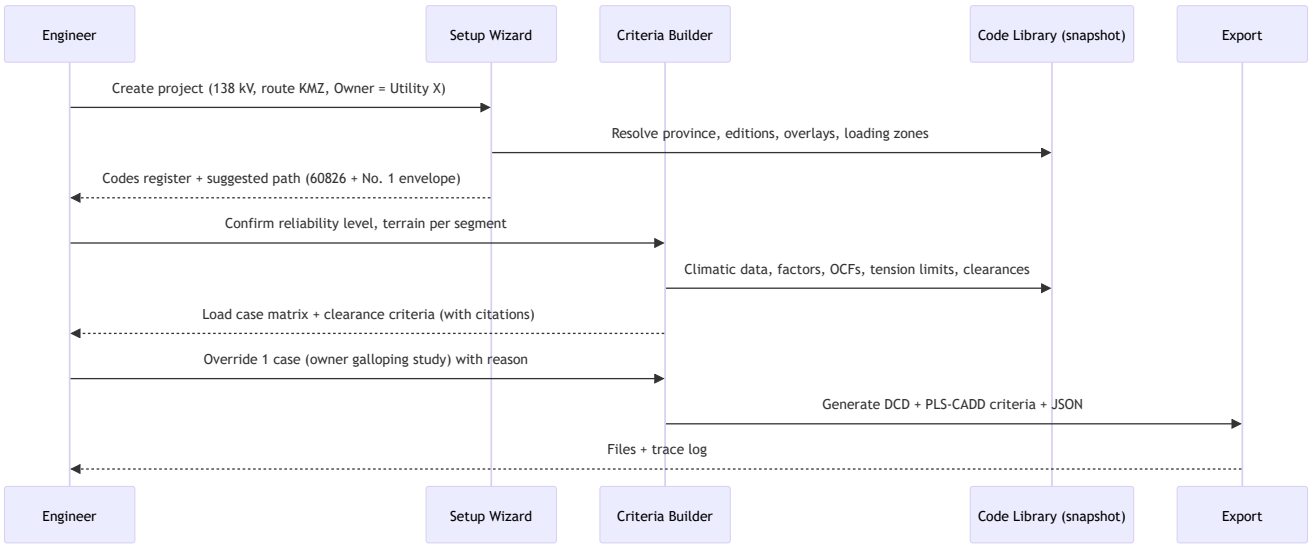
M8 — Field Mode (mobile, offline)

- Offline cache of effective clearance tables for selected owner/edition.
- Quick calculators (clearance required, pole class check, guy check).
- Measurement entry (laser rangefinder heights, temperature at time of measurement) → **temperature-corrected sag** projection to max-temp condition → pass/fail.
- Photo + GPS capture linked to structure IDs; RFI creation.

M9 — AI Code Assistant (Phase 3)

- Retrieval-augmented over licensed library only; answers include clause IDs and edition; shows retrieved passages; refuses ungrounded answers; logs every Q&A for audit.
- Example: "*What's the governing horizontal clearance for a 25 kV line beside a 3-storey building under wind displacement for Utility X?*" → answer with table/row/overlay citations and a link to open the Blowout calculator pre-filled.

9.5 Key user flow — New transmission line criteria (T1)



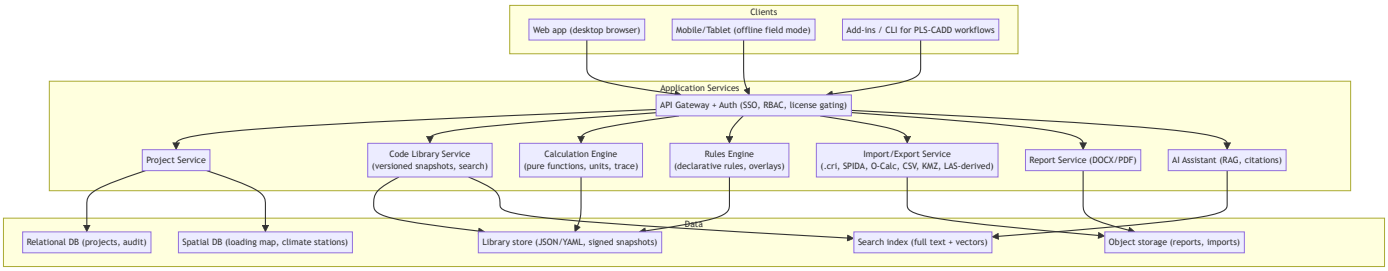
9.6 Non-functional requirements

Category	Requirement
Accuracy	Calculators match benchmark cases within stated tolerance (e.g., ≤ 0.5 % or code rounding rules)
Traceability	100 % of computed outputs have a trace record

Category	Requirement
Performance	Lookup < 200 ms; batch of 1,000 poles < 2 min; 100 km line clearance report ingestion < 1 min
Availability	Offline reference & calculators; cloud sync when online
Security	SSO (Azure AD/Entra), role-based access, license-entitlement gating of licensed text
Audit	Immutable logs of overrides, approvals, snapshot changes
Units	SI primary; imperial display; no mixed-unit storage
Accessibility	WCAG 2.2 AA for web UI
Data residency	Canadian hosting option (common utility requirement)

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10. System Architecture



10.1 Recommended technology choices (adaptable)

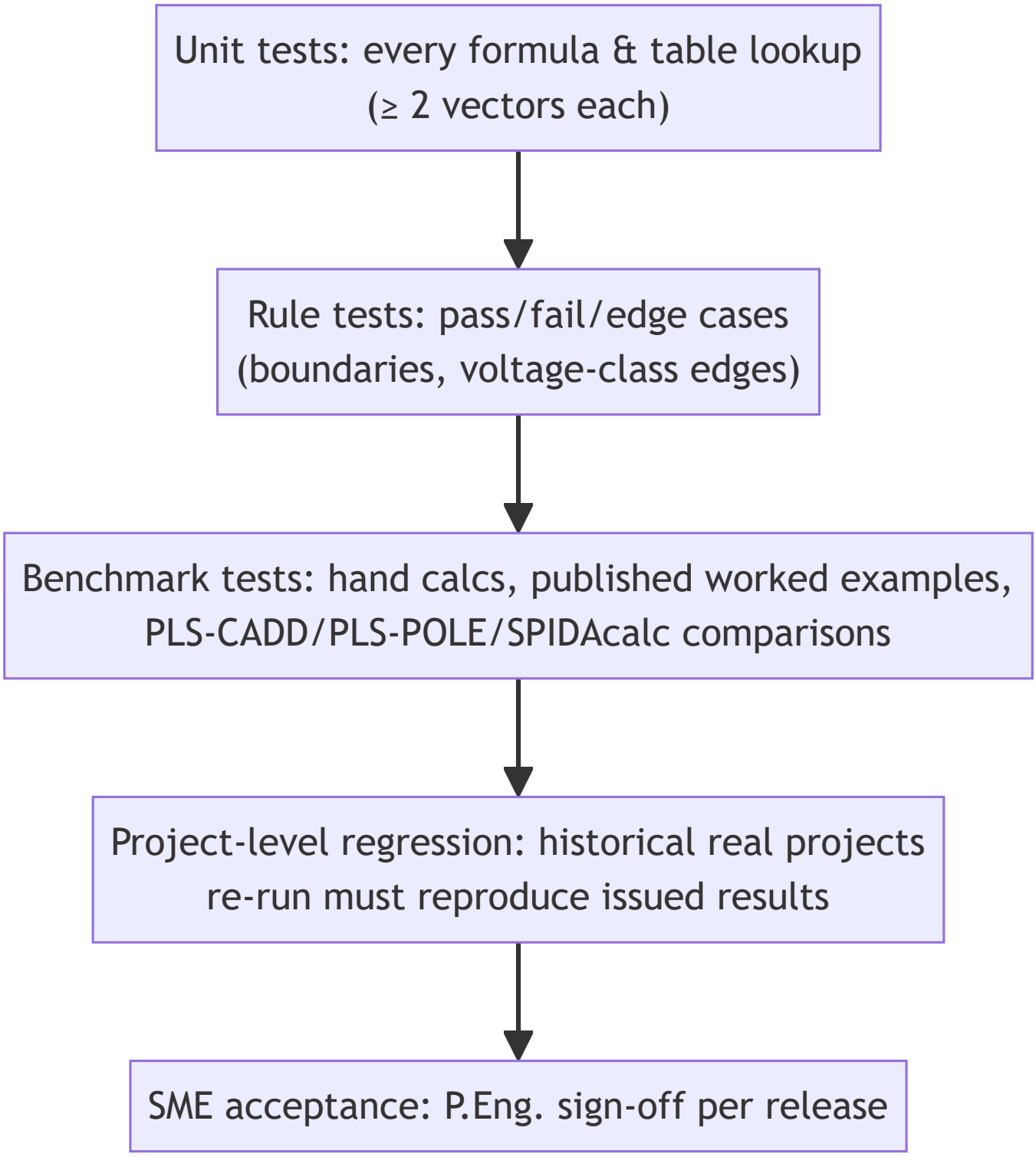
Layer	Recommendation	Rationale
Calculation core	**Single shared library** (e.g., TypeScript or Python) with a units library; pure, side-effect-free functions	Same results on web, mobile and server; easy testing
Rules	Declarative YAML/JSON rules evaluated by a small interpreter (or JSON-Logic/CEL)	SMEs can review rules without reading code
Library storage	Git-backed YAML/JSON → compiled, signed snapshot bundles	Diffable, reviewable, versioned
Spatial	PostGIS + vector tiles	Loading map and climatic station lookup
Search	Full-text + vector index (licensed text only, entitlement-filtered)	Smart search & AI grounding
Frontend	Modern web SPA + PWA offline; optional cross-platform mobile	Desk + field coverage
Reports	Templated DOCX/PDF generation	Owner templates, stampable
Hosting	Canadian region cloud or on-prem option	Utility data residency

10.2 Calculation trace record


```
{
  "calc_id": "c-2026-000123",
  "calculator": "clearance.vertical",
  "library_snapshot": "LIB-2026.03-sha256:…",
  "inputs": { "voltage_kV": 138, "surface_category": "road", "condition": "max_temp" },
  "steps": [
    { "op": "lookup", "ref": "C22.3-1:2020/T-x row r", "value": "X.XX m" },
    { "op": "adder", "ref": "R-VADDER-01", "value": "0.XX m" },
    { "op": "overlay", "ref": "OVL-UTILITYX-CLR-BUF", "value": "0.30 m" }
  ],
  "result": { "required_m": "…", "actual_m": "…", "margin_m": "…", "status": "PASS" },
  "user": "engineer@utility", "timestamp": "2026-10-12T18:20:00-05:00",
  "overrides": []
}
```

11. Verification, Validation & Quality Management

11.1 Test strategy

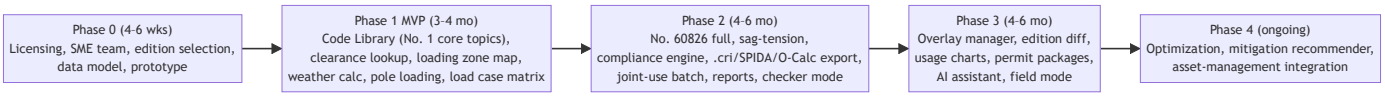


11.2 Quality controls

Control	Description
Library verification	Double-entry + SME sign-off per clause/table; signed snapshots
Software change control	Every release has release notes, test evidence and impact report
Edition change control	New edition → diff report → impacted projects list → re-validation
Override governance	Overrides require reason, are highlighted in reports, and are reviewed in Checker Mode
Independent check	Checker Mode enforces documented independent review before "Issued" status
Regulator alignment	Align with provincial guidance on use of engineering software, documented checks and authentication
Disclaimers	Visible in reports: tool version, snapshot, responsible engineer

---

12. Roadmap & Delivery Plan



12.1 MVP scope (Phase 1)

Included	Excluded (later)
C22.3 No. 1: loading conditions & map, construction grades, OCFs, vertical/horizontal clearances, building clearances, same-structure separations, tension limits	Full 60826 security/safety module
Clearance lookup calculator with owner overlay (one owner)	Multi-owner overlay manager
Weather load calculator (No. 1 + basic 60826 wind/ice)	Full return-period statistics tools
Wood pole loading calculator (CSA O15 classes)	Steel/lattice member checks (export to PLS)
Load case matrix (view + CSV export)	Direct `.cri` export
Project setup + snapshot pinning + trace	Compliance engine ingestion

12.2 Team

Role	FTE (indicative)
Product owner (senior line engineer, P.Eng.)	0.5–1.0
SME reviewers (T&D P.Eng.)	1–2 (part-time pool)
Code digitization technologists	2
Software engineers (calc core, backend, frontend)	3–4
QA / test engineer	1
UX designer	0.5

12.3 Success metrics

KPI	Target
Time to produce DCD	−50 %
Time per pole-attachment application	−60 %
Clearance-check review time per 10 km	−40 %
Post-issue non-conformances attributable to code lookup errors	→ 0
% outputs with complete trace	100 %

KPI	Target
Engineer adoption (weekly active)	> 70 % of target users at 6 months

---

13. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
CSA licensing does not permit text embedding	High	Medium	Store IDs, paraphrased intent and user-verified values; deep-link to licensed copy; negotiate enterprise license early
Digitization error in a table value	Very high	Medium	Double entry, SME sign-off, tests, signed snapshots, "VERIFIED" status gating
Wrong edition/jurisdiction applied	High	Medium	Mandatory codes register, project snapshot pinning, jurisdiction resolver
Over-reliance on automation	High	Medium	Trace UI, checker mode, overrides highlighted, training
Integration formats change (PLS, SPIDA, O-Calc)	Medium	Medium	Adapter layer, versioned parsers, vendor partnerships
AI hallucination	High	Medium	RAG-only, citation required, refusal policy, logging, disabled for normative decisions
Scope creep into full FEA/structural analysis	Medium	High	Clear boundary: generate inputs/check outputs of established tools
Loading-zone boundary ambiguity	Medium	Medium	Boundary buffer + conservative default + flag

14. Appendices

Appendix A — Glossary

Term	Meaning
AHJ	Authority having jurisdiction
Blowout	Horizontal conductor displacement under wind
Construction grade	C22.3 No. 1 classification setting strength requirements (e.g., Grade 1 / Grade 2)
DCD	Design Criteria Document (basis of design)
Envelope approach	Designing for the governing result of multiple code paths
H/w	Catenary parameter (horizontal tension / unit weight) used for vibration control
OCF	Overload Capacity Factor (deterministic load multiplier)
OPGW / OHGW	Optical ground wire / overhead ground wire
RBD	Reliability-based design (C22.3 No. 60826)
RSL	Residual static load (security case)
RTS	Rated tensile strength
Ruling span	Equivalent span representing a tension section
Snapshot	Immutable, signed version of the code library + overlays pinned to a project
Usage chart	Allowable wind span / weight span vs line angle diagram for a structure type

Appendix B — Design Criteria Document (DCD) template (auto-generated sections)

1. Project description (facility, voltage, circuits, route, owner)
2. Applicable codes, standards & regulations (editions, amendments, overlays)
3. Design path (No. 1 / No. 60826 / envelope) and justification
4. Reliability level & return periods (if 60826)
5. Climatic data (sources, stations, interpolation method, altitude/terrain adjustments)
6. Loading zone(s) and deterministic loading conditions
7. Load case matrix (climatic, security, safety, construction, deterministic)
8. Load / overload capacity factors and strength factors by component
9. Conductor & shield wire data; tension limits; vibration criteria; creep assumptions
10. Clearance criteria table (base + adders + buffers) and governing conditions
11. Separation criteria (structure & midspan), galloping criteria
12. Insulator swing cases and air gaps
13. Structure families, materials and standards (CSA O15, S16, etc.)
14. Guying, anchoring and foundation criteria
15. Grounding & bonding criteria
16. Joint-use criteria

17. Special crossings & permits (rail, navigable water, highways, pipelines, aviation)
18. Deviations, overrides & interpretations (with justification)
19. Revision history & authentication block

Appendix C — Phase checklists (condensed)

Phase 1 — Criteria

- [ ] Adopted edition(s) confirmed for jurisdiction
- [ ] Owner overlay version recorded
- [ ] Loading zone(s) verified incl. boundary proximity
- [ ] Climatic sources documented; return period conversion recorded
- [ ] Terrain categories assigned per segment
- [ ] Construction grade(s) assigned (crossings, voltage, location)
- [ ] Tension limits and vibration criteria set
- [ ] Clearance criteria table with buffers issued
- [ ] Security & C&M cases defined (60826)
- [ ] Criteria exported and import into PLS-CADD verified

Phase 4 — Spotting & clearances

- [ ] All crossings in register modeled with correct feature codes
- [ ] Max-temp, ice-loaded and blowout conditions checked
- [ ] Midspan and crossing separations checked
- [ ] Marginal items (within buffer) reviewed
- [ ] Clearance report generated with citations

Phase 5/6 — Structures, guys, foundations

- [ ] All load cases applied to each structure type
- [ ] Usage charts confirm every structure within limits
- [ ] Insulator swing clearances verified
- [ ] Guys/anchors sized and guy insulators/guards per code
- [ ] Foundations designed with geotechnical inputs

Phase 8 — QA

- [ ] Independent check completed in Checker Mode
- [ ] Overrides reviewed and justified
- [ ] Library snapshot & software version recorded
- [ ] Authentication (stamp) applied

Appendix D — Sample rules catalogue (initial set)

Rule ID	Topic	Description
R-CLR-VERT-001	clearance.vertical	Conductor-to-ground/feature vertical clearance
R-CLR-HORZ-001	clearance.horizontal	Wind-displaced conductor to building/structure
R-CLR-POOL-001	clearance.pool	Clearances over/near swimming pools
R-SEP-STR-001	separation.same_structure	Phase/circuit/supply–comm separation at structure
R-SEP-MID-001	separation.midspan	Sag-dependent midspan separation
R-SEP-XING-001	separation.crossing	Line-crossing vertical separation
R-TEN-001	conductor.tension_limit	Max tension per condition vs % RTS
R-STR-WOOD-001	strength.wood	Wood pole groundline utilization with OCF
R-STR-GUY-001	strength.guy	Guy tension with OCF vs rated strength
R-INS-001	strength.insulator	Insulator mechanical load vs rating factor
R-JU-001	jointuse	Supply–comm separation & climbing space
R-RBD-001	loading.*	$\gamma_U \cdot Q_T \leq \Phi_R \cdot R_c$ component check

Appendix E — Items requiring SME verification before production

#	Item	Source to verify
1	Adopted edition of C22.3 No. 1 and No. 60826 per province/territory	Provincial regulations / AHJ
2	Loading conditions table values (all zones) and loading map boundaries	C22.3 No. 1 (adopted edition)
3	Construction grade criteria and OCF tables	C22.3 No. 1
4	All clearance and separation tables, voltage adders, sag conditions	C22.3 No. 1
5	Conductor tension limits	C22.3 No. 1
6	Reliability levels / return periods and Canadian deviations	C22.3 No. 60826
7	Canadian climatic data annex values and conversion factors	C22.3 No. 60826
8	Wind factors (K_R, G_c, G_L, G_t, C_x), ice combination factors	C22.3 No. 60826
9	Strength factors (Φ_N, Φ_Q, Φ_S, Φ_C) and coordination rules	C22.3 No. 60826
10	Wood pole fibre stresses, dimensions, classes	CSA O15
11	Insulator & hardware rating factors	C22.3 No. 1, CSA C411.1, CSA C83
12	Owner overlays (buffers, stricter values)	Utility standards

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End of document.